TSM 098 | INDEPENDENT PRACTICE

Linear equations and checking systems

While TSM 099 studies Topic 32. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 31.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: two-term and short linear equations

1. Solve and check in the original equation.

$$-6x - 2 = 16$$

2. Solve and check in the original equation.

$$-3x - 1 = -7$$

3. Solve and check in the original equation.

$$-6x - 7 = 17$$

4. Solve and check in the original equation.

$$-3x + 8 = 8$$

5. Solve and check in the original equation.

$$-6x + 8 = -28$$

6. Solve and check in the original equation.

$$2x + 3 = 11$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: variable terms on both sides

7. Solve and check in the original equation.

$$-5x + 1 = x - 8$$

8. Solve and check in the original equation.

$$4x + 6 = x + 1$$

9. Solve and check in the original equation.

$$-3x - 6 = 5x + 2$$

10. Solve and check in the original equation.

$$-5x + 1 = -2x - 5$$

11. Solve and check in the original equation.

$$-3x - 7 = 5x + 5$$

12. Solve and check in the original equation.

$$-5x + 1 = -2x + 3$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: review linear equations with fractions

13. Solve and check in the original equation.

$$(\frac{x-2}{2})+(\frac{x-4}{3})=(\frac{-2x-5}{6})$$

14. Solve and check in the original equation.

$$(\frac{3x-5}{2})+(\frac{x-1}{6})=(\frac{-3}{3})$$

15. Solve and check in the original equation.

$$(\frac{-3x+3}{2})+(\frac{x-1}{5})=(\frac{-x+1}{8})$$

16. Solve and check in the original equation.

$$(\frac{x-4}{4})+(\frac{2}{3})=(\frac{0}{8})$$

17. Solve and check in the original equation.

$$(\frac{-3x-3}{8})+(\frac{x-2}{2})=(\frac{-2x-6}{5})$$

18. Solve and check in the original equation.

$$(\frac{3x-1}{3})+(\frac{2}{4})=(\frac{-2x-3}{2})$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: check proposed solutions of systems

19. Does the ordered pair solve both equations? Show both substitutions.

$(2, 3): \quad x + y = 5, \quad 2x - y = 1$

20. Does the ordered pair solve both equations? Show both substitutions.

$(3, -1): \quad x - y = 4, \quad 2x + y = 8$

21. Does the ordered pair solve both equations? Show both substitutions.

$(2, -1): \quad 2x + 3y = 1, \quad x - 2y = 4$

22. Does the ordered pair solve both equations? Show both substitutions.

$(1, 3): \quad x + 2y = 7, \quad 3x - y = 0$

23. Does the ordered pair solve both equations? Show both substitutions.

$(3, 1): \quad 2x - y = 5, \quad x + y = 4$

24. Does the ordered pair solve both equations? Show both substitutions.

$(-1, 1): \quad 3x + y = -2, \quad x - y = -2$

TSM 098 | ANSWER KEY

Linear equations and checking systems

While TSM 099 studies Topic 32. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 31.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: two-term and short linear equations

1. Solve and check in the original equation.

$$-6x - 2 = 16$$

$x = -3$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$-3x - 1 = -7$$

$x = 2$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-6x - 7 = 17$$

$x = -4$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$-3x + 8 = 8$$

$x = 0$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$-6x + 8 = -28$$

$x = 6$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$2x + 3 = 11$$

$x = 4$

Check by substituting into the original equation.

Linear equations and checking systems

While TSM 099 studies Topic 32. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 31.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: variable terms on both sides

7. Solve and check in the original equation.

$-5x + 1 = x - 8$

$x = \frac{3}{2}$

Check by substituting into the original equation.

8. Solve and check in the original equation.

$4x + 6 = x + 1$

$x = -\frac{5}{3}$

Check by substituting into the original equation.

9. Solve and check in the original equation.

$-3x - 6 = 5x + 2$

$x = -1$

Check by substituting into the original equation.

10. Solve and check in the original equation.

$-5x + 1 = -2x - 5$

$x = 2$

Check by substituting into the original equation.

11. Solve and check in the original equation.

$-3x - 7 = 5x + 5$

$x = -\frac{3}{2}$

Check by substituting into the original equation.

12. Solve and check in the original equation.

$-5x + 1 = -2x + 3$

$x = -\frac{2}{3}$

Check by substituting into the original equation.

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: review linear equations with fractions

13. Solve and check in the original equation.

$$\frac{5x-14}{6}=\frac{-2x-5}{6}$$

$$6(5x-14)=6(-2x-5)$$

$$x=\frac{9}{7}$$

Check by substituting into the original equation.

14. Solve and check in the original equation.

$$\frac{5x-8}{3}=-1$$

$$1(5x-8)=3(-1)$$

$$x=1$$

Check by substituting into the original equation.

15. Solve and check in the original equation.

$$\frac{-13x+13}{10}=\frac{-x+1}{8}$$

$$8(-13x+13)=10(-x+1)$$

$$x=1$$

Check by substituting into the original equation.

16. Solve and check in the original equation.

$$\frac{3x-4}{12}=0$$

$$1(3x-4)=12(0)$$

$$x=\frac{4}{3}$$

Check by substituting into the original equation.

17. Solve and check in the original equation.

$$\frac{x-11}{8}=\frac{-2x-6}{5}$$

$$5(x-11)=8(-2x-6)$$

$$x=\frac{1}{3}$$

Check by substituting into the original equation.

18. Solve and check in the original equation.

$$\frac{6x+1}{6}=\frac{-2x-3}{2}$$

$$2(6x+1)=6(-2x-3)$$

$$x=-\frac{5}{6}$$

Check by substituting into the original equation.

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: check proposed solutions of systems

19. Does the ordered pair solve both equations? Show both substitutions.

$1(2)+(1)(3)=5;\quad 2(2)+(-1)(3)=1$

Yes; both equations are true.

20. Does the ordered pair solve both equations? Show both substitutions.

$1(3)+(-1)(-1)=4;\quad 2(3)+(1)(-1)=5$

No; at least one equation is false.

21. Does the ordered pair solve both equations? Show both substitutions.

$2(2)+(3)(-1)=1;\quad 1(2)+(-2)(-1)=4$

Yes; both equations are true.

22. Does the ordered pair solve both equations? Show both substitutions.

$1(1)+(2)(3)=7;\quad 3(1)+(-1)(3)=0$

Yes; both equations are true.

23. Does the ordered pair solve both equations? Show both substitutions.

$2(3)+(-1)(1)=5;\quad 1(3)+(1)(1)=4$

Yes; both equations are true.

24. Does the ordered pair solve both equations? Show both substitutions.

$3(-1)+(1)(1)=-2;\quad 1(-1)+(-1)(1)=-2$

Yes; both equations are true.